

Nekhil Nagia

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EXPERIENCE

Amazon Web Services SDE1

August 2021 – Present | Toronto

- Working on the [RDS Aurora Serverless Control Plane](#) team responsible for managing and developing systems powering one of the fastest growing services in AWS
- [Optimized control plane components by 2x](#) to enable scale down to ultra-low resource instances
- Setup distributed infrastructure for billing customers based on their serverless usage and working on internal APIs handling [100s of requests per second](#)
- Actively involved in code reviews, design reviews and operations of the control plane

Amazon Web Services SDE Intern

May 2020 – August 2020 | Toronto

- Worked on the [RDS Aurora Control Plane](#) team responsible for managing and developing control plane components on all provisioned RDS Aurora clusters
- Improved Aurora QA for [availability](#) by adding over [60 failure injection and additional stress tests](#)
- Successfully integrated work with release pipelines thus ensuring Aurora availability SLAs and prevent regressions during new releases

Square Emerging Machine Learning Developer

Feb 2020 – May 2020 | Toronto

- Worked on the [Risk Data Science](#) team to develop solutions to mitigate fraud in Cash App
- Worked on an end-to-end deep learning pipeline to predict chargebacks using raw event level customer data, used tools such as [TensorFlow](#), [SQL](#), [Apache Airflow](#)
- Built an [AI receptionist](#) system called Deep Doorman David as a [side project](#) that uses a custom [face recognition](#) model to interact with visitors at the office and notifies the appropriate employees on Slack

Dessa Software Engineering Intern

May 2019 – Feb 2020 | Toronto

- Building a product called [Foundations](#) for managing [end-to-end machine learning](#) lifecycle
- Involved in all areas of engineering, from setting up devops infrastructure in Jenkins & AWS to architecture and development using tools such as [Kubernetes](#), [Redis](#) and [Python](#)
- Worked on a feature to allow the user to automatically track [data health](#) and consistency using intelligent data validation techniques and alert them to prevent model degradation

PROJECTS

Federated Learning for Wi-Fi Fingerprinting

2021 | Toronto

- Worked under the supervision of a PhD student and Professor Valaee at [WIRLab UofT](#)
- Created a [novel application of federated learning](#) for the Wi-Fi fingerprinting problem and demonstrating performance gains in localization accuracy
- Project culminated with a [research paper](#) that is currently under review at a [top conference](#) in the field of wireless communications

ACTIVITIES

Meditation, Camping & Hiking, Beginner pilot, Travelling

EDUCATION

UNIVERSITY OF TORONTO Bachelor of Applied Science in Computer Engineering

Graduated June 2021